

▼ Numpy and Pandas

Pandas

```
import pandas as pd
import numpy as np
import seaborn as sns

df = sns.load_dataset('titanic')
df.head()
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

```
df2 = df.copy()

print('Before dropping: ', df.shape)
df_dropped = df.dropna()
print('After dropping: ', df_dropped.shape)
```

```
Before dropping: (891, 15)
After dropping: (182, 15)
```

```
df2['age'] = df2['age'].fillna(df2['age'].mean())
df2['embarked'] = df2['embarked'].fillna(df2['embarked'].mode()[0])

df2 = df2.drop(columns=['deck'])
print('Missing after cleaning')
print(df2.isnull().sum())
```

```
Missing after cleaning
survived      0
pclass        0
sex           0
age           0
sibsp         0
parch         0
fare          0
embarked      0
class         0
who           0
adult_male    0
embark_town   2
alive         0
```

```
alone      0
dtype: int64
```

```
output_file = 'titanic_data.xlsx'
df2.to_excel(output_file, index = False )
print(f"Dataset saved to {output_file}")
```

Dataset saved to titanic_data.xlsx

```
survivors_young = df2.query('survived == 1 and age < 18')
print(len(survivors_young))
```

61

Double-click (or enter) to edit

```
# Find all male passengers who paid a fare above 200 and survived.
# df_male = df2.filter(df2['sex'] == 'male').filter(df2['fare'] > 200).filter(df2['alive'] == 'yes')
df_male = df2.query("sex == 'male' and fare > 200 and alive == 'yes'")
print(df_male)
```

```
#Using iloc, extract 100 to 109 rows and the last 3 columns
df2.iloc[100:110, -4:-1]
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
679	1	1	male	36.0	0	1	512.3292	C	First	
737	1	1	male	35.0	0	0	512.3292	C	First	

	who	adult_male	embark_town	alive	alone
679	man	True	Cherbourg	yes	False
737	man	True	Cherbourg	yes	True

	adult_male	embark_town	alive
100	False	Southampton	no
101	True	Southampton	no
102	True	Southampton	no
103	True	Southampton	no
104	True	Southampton	no
105	True	Southampton	no
106	False	Southampton	yes
107	True	Southampton	yes
108	True	Southampton	no
109	False	Queenstown	yes

```
# pivot table - Like excel
```

```
pt = df2.pivot_table(
    values = 'survived',
    index = 'pclass',
    columns = 'sex',
```

```
aggfunc= 'mean'
)

pt
```

	sex	female	male
pclass			
1		0.968085	0.368852
2		0.921053	0.157407
3		0.500000	0.135447

```
#Group By and Aggregation
```

```
survival_stats = (
    df2.groupby(['pclass', 'sex'])['survived']
    .agg(['mean', 'count'])
    .rename(columns = {'mean': 'survival_rate', 'count': 'total_count'})
    .sort_values(by='survival_rate', ascending=False)
)

survival_stats
```

		survival_rate	total_count
pclass	sex		
1	female	0.968085	94
2	female	0.921053	76
3	female	0.500000	144
1	male	0.368852	122
2	male	0.157407	108
3	male	0.135447	347

```
# Multiple Aggregations on different columns at once
```

```
class_summery = df2.groupby('pclass').agg(
    avg_age = ('age', 'mean'),
    avg_fare = ('fare', 'mean'),
    survived = ('survived', 'count'),
    n = ('survived', 'sum') # Sum will incorporate 0
)

class_summery
```

	avg_age	avg_fare	survived	n
pclass				
1	37.048118	84.154687	216	136
2	29.866958	20.662183	184	87
3	26.403259	13.675550	491	119

```
# Using groupBy find :
#Average fare paid by passengers from each embarkation port
# df2.head()
average_fare = (
    df2.groupby(['embarked'])['fare']
    .agg(['mean'])
    .rename(columns = {'mean' : 'Average Fare'})
)
average_fare

#The number of survivors per class sorted from highest to lowest
no_of_survivors = (
    df2.groupby(['pclass'])['survived']
    .agg(['sum'])
    .rename(columns={'sum' : 'Survivors Count'})
    .sort_values(by='Survivors Count', ascending=False)
)
no_of_survivors
```

	Survivors Count
pclass	
1	136
3	119
2	87

```
#Merging DataFrames
```

```
#Look up tables

class_info = pd.DataFrame({
    'pclass': [1,2,3],
    'class_name' : ['First', 'Second', 'Third']
})

class_info
```

```
#Create a second lookup table with port full names

port_info = pd.DataFrame({
    'embarked' : ['C', 'Q', 'S'],
    'port_name' : ['Cherbourg', 'Queenstown', 'Southampton']
})
```

```
port_info

#C - Cherbourg, Q - Queenstown, S - Southampton
#Merge it with df2 on 'embarked' and confirm the join worked

df2 = pd.merge(df2, port_info, on = 'embarked', how = 'left')
df2.head() # Validation
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	embark_town	alive	alone	port_name
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	Southampton	no	False	Southampton
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	Cherbourg	yes	False	Cherbourg
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	Southampton	yes	True	Southampton
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	Southampton	yes	False	Southampton
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	Southampton	no	True	Southampton

```
# Feature Engineering : age group

df2['age_group'] = pd.cut(
    df2['age'],
    bins = [0, 18, 50, 100],
    labels = ['Child', 'Adult', 'Senior']
)

df2['age_group'].unique()

['Adult', 'Senior', 'Child']
Categories (3, object): ['Child' < 'Adult' < 'Senior']
```